

RETAIL SALES DATA ANALYSIS USING PYTHON

1. Introduction

Retail businesses generate large amounts of transactional data every day. Analyzing this data can help organizations understand customer behaviour, product performance, sales patterns, profitability, and regional performance.

This project performs an end-to-end analysis of the Sample Superstore retail dataset using Python.

Python libraries including NumPy, Pandas, Matplotlib, and Seaborn were used for data processing, analysis, and visualization.

2. Project Objective

The objective of this project is to analyze retail sales data and generate meaningful business insights.

The main objectives are:

- Clean and preprocess the retail sales dataset
- Perform descriptive statistical analysis
- Analyze sales and profitability
- Identify the highest revenue-generating products
- Analyze regional sales performance
- Identify high-value customers
- Analyze monthly sales trends
- Identify loss-making sub-categories
- Study relationships between numerical variables
- Create visualizations for business understanding
- Generate actionable business insights

3. Dataset Description

The Sample Superstore dataset was used for this project.

The dataset contains retail transaction information including:

- Order ID
- Order Date
- Ship Date
- Customer ID
- Customer Name
- Segment
- Country
- City
- State
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

4. Tools and Technologies

The following technologies were used:

Programming Language

Python

Libraries

- NumPy
- Pandas

- Matplotlib
- Seaborn

Development Environment

Jupyter Notebook using Anaconda

Version Control

Git and GitHub

5. Data Loading

The dataset was imported into Python using the Pandas library.

Pandas was used to inspect rows, columns, data types, descriptive statistics, and general dataset information.

6. Data Cleaning

Before performing the analysis, the dataset was checked for data quality issues.

The following steps were performed:

- Missing values were checked
- Duplicate records were checked
- Duplicate rows were removed where necessary
- Column data types were verified
- Order Date was converted into datetime format
- Additional year and month variables were created
- Text columns were inspected
- Numerical variables were checked for unusual values

A cleaned dataset was prepared for further analysis.

7. NumPy Analysis

NumPy was used to perform numerical calculations on the Sales and Profit columns.

The calculations included:

- Total Sales
- Average Sales
- Median Sales
- Minimum Sales
- Maximum Sales
- Standard Deviation
- Total Profit
- Average Profit
- Median Profit

This demonstrated the use of NumPy arrays and mathematical operations for data analysis.

8. Descriptive Statistics

Descriptive statistics were calculated for:

- Sales
- Quantity
- Discount
- Profit

Measures such as mean, median, mode, variance, and standard deviation were used to understand the distribution of the numerical variables.

9. Category Analysis

The three main categories were analyzed based on sales, profit, and quantity.

Furniture

Sales: 741,999.80

Profit: 18,451.27

Quantity: 8,028

Office Supplies

Sales: 719,047.03

Profit: 122,490.80

Quantity: 22,906

Technology

Sales: 836,154.03

Profit: 145,454.95

Quantity: 6,939

Technology generated the highest sales and profit.

Furniture generated substantial sales but comparatively low profitability.

Office Supplies had the highest total quantity sold.

10. Product Analysis

Products were grouped according to total sales.

The highest revenue-generating product was:

Canon imageCLASS 2200 Advanced Copier

Sales: 61,599.82

Other high-value products included:

- Fellowes PB500 Electric Punch Plastic Comb Binding Machine
- Cisco TelePresence System EX90 Videoconferencing Unit
- HON 5400 Series Task Chairs for Big and Tall
- GBC DocuBind TL300 Electric Binding System

The results show that premium technology and business equipment products contribute significantly to revenue.

11. Regional Analysis

Regional performance was analyzed based on sales, profit, and number of orders.

West

Sales: 725,457.82

Profit: 108,418.45

Orders: 1,611

East

Sales: 678,781.24

Profit: 91,522.78

Orders: 1,401

Central

Sales: 501,239.89

Profit: 39,706.36

Orders: 1,175

South

Sales: 391,721.91

Profit: 46,749.43

Orders: 822

The West region generated the highest sales and profit.

The East region was the second strongest performing region.

Although Central generated more sales than South, South achieved higher profit.

12. Monthly Sales Trend Analysis

Sales were grouped by month to identify changes over time.

The highest monthly sales occurred in:

November 2017

Sales: 118,447.83

The lowest monthly sales occurred in:

February 2014

Sales: 4,519.89

The monthly trend analysis helps identify seasonal patterns and periods of strong business performance.

13. Customer Analysis

Customers were grouped according to total sales.

The highest revenue-generating customer was:

Sean Miller

Sales: 25,043.05

Other top customers included:

- Tamara Chand
- Raymond Buch
- Tom Ashbrook
- Adrian Barton
- Ken Lonsdale
- Sanjit Chand
- Hunter Lopez
- Sanjit Engle
- Christopher Conant

Identifying high-value customers can help businesses develop customer retention and loyalty strategies.

14. Underperforming Sub-Categories

Sub-categories were analyzed according to total profit.

Three sub-categories generated overall losses:

Tables

Profit: -17,725.48

Bookcases

Profit: -3,472.56

Supplies

Profit: -1,189.10

Tables represented the largest loss-making sub-category.

These categories should be reviewed for factors such as discounting, pricing, shipping costs, and product margins.

15. Correlation Analysis

Correlation analysis was performed between:

- Sales
- Quantity
- Discount
- Profit

Important results included:

Sales and Profit correlation: approximately 0.48

This indicates a moderate positive relationship between sales and profit.

Discount and Profit correlation: approximately -0.22

This indicates a negative relationship between discounts and profitability.

Quantity and Profit correlation: approximately 0.07

This relationship is very weak.

Quantity and Sales correlation: approximately 0.20

This indicates a weak positive relationship.

16. Data Visualization

The following visualizations were created:

- Sales by Category Bar Chart
- Sales by Region Bar Chart
- Profit by Region Bar Chart
- Top Products Bar Chart

- Top Customers Bar Chart
- Monthly Sales Trend Line Chart
- Profit by Sub-Category Bar Chart
- Sales Distribution Histogram
- Profit Distribution Histogram
- Correlation Heatmap
- Sales Box Plot

These visualizations make the business results easier to understand and communicate.

17. Key Business Insights

The analysis generated the following important insights:

- Technology is the strongest category based on both sales and profit.
- The West region is the highest-performing region.
- Canon imageCLASS 2200 Advanced Copier is the highest revenue-generating product.
- Sean Miller is the highest revenue-generating customer.
- November 2017 recorded the highest monthly sales.
- February 2014 recorded the lowest monthly sales.
- Tables are the largest loss-making sub-category.
- Sales have a moderate positive relationship with profit.
- Discounts have a negative relationship with profit.
- High sales do not always guarantee high profitability.

18. Recommendations

Based on the analysis:

- Focus on high-performing Technology products.
- Strengthen customer retention strategies for high-value customers.
- Investigate the poor profitability of Tables, Bookcases, and Supplies.
- Review discount strategies because higher discounts may negatively affect profit.

- Study the strategies used in the West region and apply successful approaches to weaker regions.
- Prepare inventory and marketing campaigns before historically high-sales periods.

19. Conclusion

The Retail Sales Data Analysis project successfully demonstrates the use of Python for business data analysis.

The project included data cleaning, preprocessing, NumPy operations, statistical analysis, exploratory data analysis, visualization, and business insight generation.

The analysis identified important trends related to products, customers, categories, regions, sales, discounts, and profitability.

The project demonstrates how data analytics can help businesses make informed, data-driven decisions.

20. Future Scope

Future enhancements may include:

- Interactive Streamlit dashboard
- SQL database integration
- Power BI or Tableau dashboards
- Customer segmentation
- Sales forecasting
- Machine learning models
- Cloud deployment